

Basic Principles of Medicine 1

Module: Cardiovascular, Respiratory and Renal Systems | DLA 3

Aortic Arches

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Session ID:

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Recommended reading

- Moore, Persaud, Torchia. The Developing Human: Clinically Oriented Embryology, 11th Ed
- Printed textbook: Chapter 13, Pgs. 297 – 303
- Online textbook: Chapter 13, Derivatives of Pharyngeal Arch Arteries.

[https://www-clinicalkey-com.sgu.idm.oclc.org/#!/content/book/3-s2.0-B9780323611541000138?scrollTo=%23hl0001632:~:text=C.%2C%20Boston%2C%20MA.\)-,Derivatives%20of%20Pharyngeal%20Arch%20Arteries,-7](https://www-clinicalkey-com.sgu.idm.oclc.org/#!/content/book/3-s2.0-B9780323611541000138?scrollTo=%23hl0001632:~:text=C.%2C%20Boston%2C%20MA.)-,Derivatives%20of%20Pharyngeal%20Arch%20Arteries,-7)

- Please note that prior knowledge of the normal anatomy of the great vessels and heart is needed before reviewing this material
- Pay special attention to the **Blue/Green** boxes for the clinical correlations for this topic.

Objectives

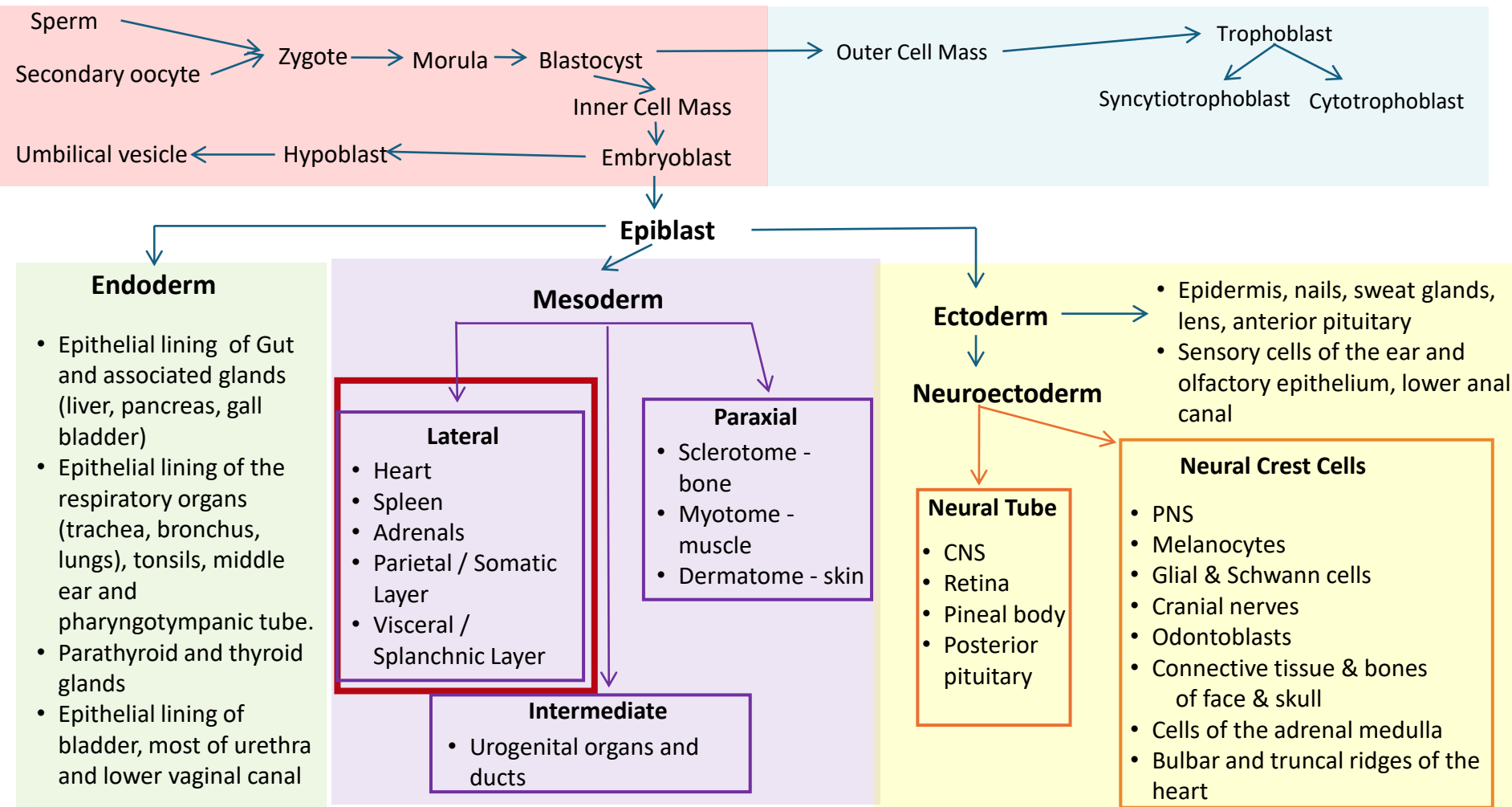
SOM.MK.II.BPM1.3.CPR.2.ANAT.1346	Summarize the major events in the transformation of the pharyngeal arch system into the adult arterial pattern.
SOM.MK.II.BPM1.3.CPR.2.ANAT.1347	Outline the embryologic basis of double aortic arch, right aortic arch, and anomalous right subclavian artery
SOM.MK.II.BPM1.3.CPR.2.ANAT.1348	Discuss the relatively common aortic arch anomalies, including patent ductus arteriosus and coarctation of the aorta.



Required Video: Development of the Vascular System

- Please review the animations associated with this DLA.
- This video provides a summary of the development of the vascular system and the aortic arches.

The Big picture



The journey of development

- Fertilization to gastrulation - **Weeks 1-3**

Organogenesis – Weeks 4 to 8

Everything happens synchronously with many simultaneous events

- Cardiac progenitor cells migrate from the cranial part of the primitive streak
 - First heart field develops in the splanchnic mesoderm
 - Consists of left and right portions (limbs)
 - Bilateral dorsal aorta develop on either side of the developing notochord
 - Lateral folding of the embryo brings the heart fields together
 - Two endocardial tubes form; one in each heart field
 - Formation of single primordial heart tube by fusion of the endocardial tubes
 - Embryo continues to fold laterally and craniocaudally
 - First pair of aortic arch arteries are formed – day 22 - 24
 - Additional pairs of aortic arch arteries added as pharyngeal arches develop
 - Aortic arch arteries and dorsal aorta remodel and give rise to great vessels of the heart, arteries of the head and neck, and descending and abdominal aorta
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- Week 3**
- Week 4**
- Weeks 5 - 7**

Special note

Some slides have a green background, this will not be covered in the video recording but are put so you have the information available

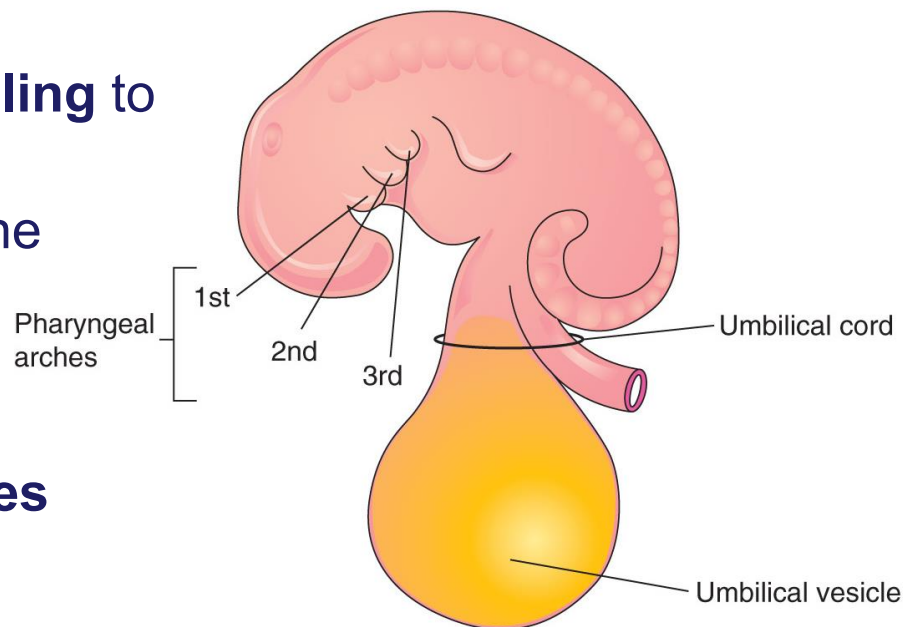


Terminology

- **Vasculogenesis** – de novo blood vessel development
- **Angiogenesis** – budding and sprouting of new vessels from existing ones
- **Cardiogenic tissue** – mesenchyme that will form the heart and blood vessels
- **Aortic arch artery/pharyngeal arch artery**: artery that connects the aortic sac and dorsal aorta during development, corresponds to the pharyngeal arches
- **Pharyngeal arch** – mesenchymal condensation that gives rise to the muscles, connective tissue, vasculature and nerve of the face, head and neck

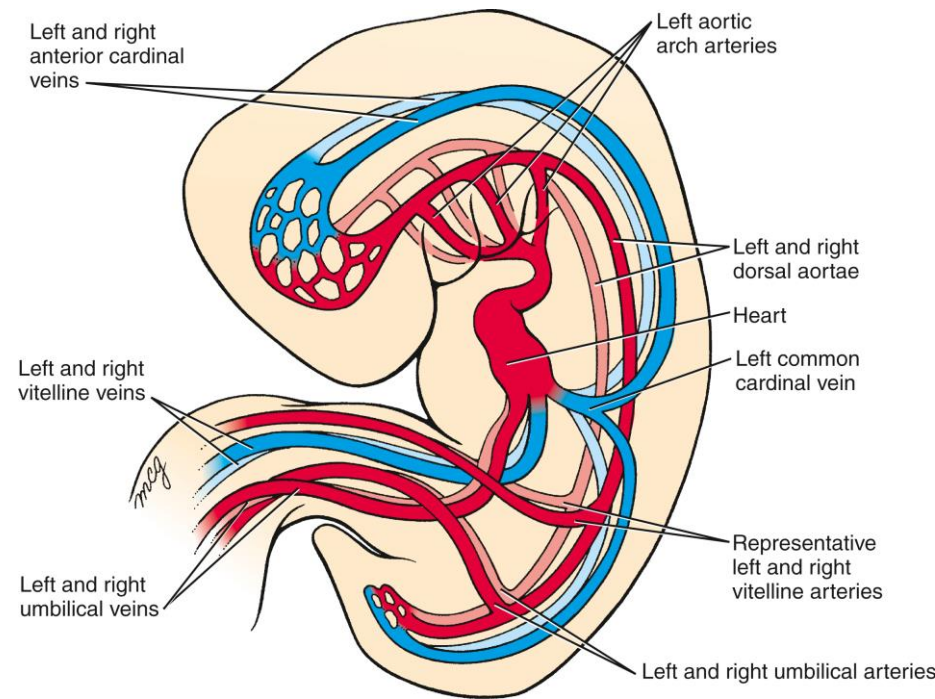
Week 4 - Pharyngeal arches

- Bilateral swellings (**mesenchymal condensations**) on the ventrolateral pharynx
 - The pharyngeal arches contain tissue from **all the germ layers**
 - Just caudal to the developing mouth
 - In fish corresponds to the development of the gills
- Separated by **ectodermal clefts** and **endodermal pouches**
- **4-6 pairs** develop in sequence from cranial to caudal
 - 5th degenerates as soon as it forms
 - Not all present simultaneously (cranial ones regress before caudal ones are fully formed)
 - **Regression is due to remodeling** to form future adult structures
- Responsible for the formation of the face and neck
- Each arch has its own **nerve, cartilage and artery**
 - Arteries are **Aortic arch arteries**



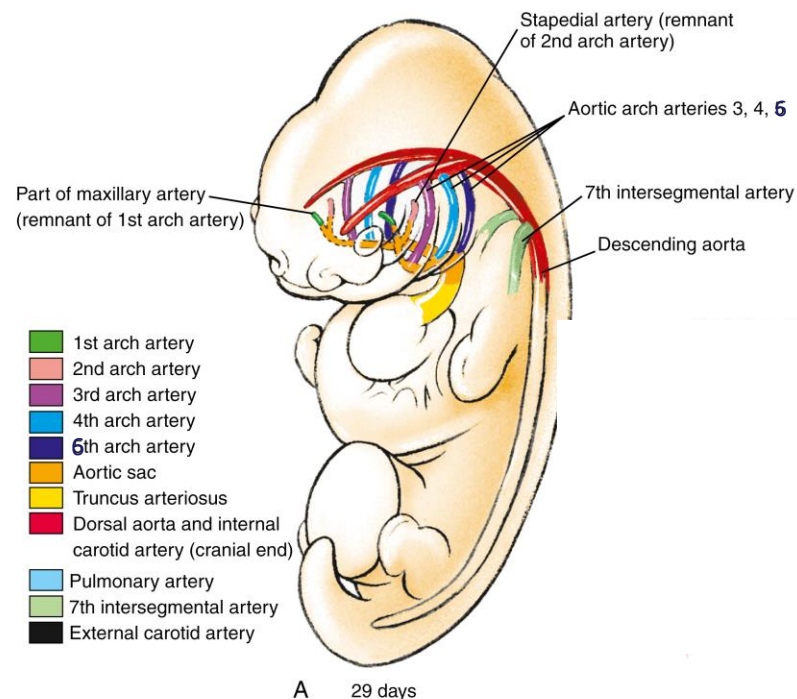
Week 4 – day 22-24 – development of the 1st aortic arch arteries

- **Lateral folding** of the embryo brings the **endocardial tubes** (future heart) and the surrounding cardiogenic tissue together
- **Head fold** brings the developing heart tube into the chest
- Paired dorsal aorta is pulled with the developing heart tube
 - Forms a dorsoventral loop = **1st aortic arch arteries**
- Connect the aortic sac to the dorsal aortae
- Located in the mesenchyme of the **1st pharyngeal arch**



Week 4 & 5 – development of additional aortic arch arteries

- Develop in a **craniocaudal** fashion
- Corresponds to the pharyngeal arches
- Arise ventrally from the **aortic sac**
- Connect dorsally to the **dorsal aortae** bilaterally
- 1-3 form bilaterally symmetrical
- 4&6 forms asymmetrically
- 2nd forms on day 26
 - Simultaneous regression of 1st
- 3rd forms on day 28
- 6th forms on day 29
 - Simultaneous regression of 2nd



Regression does not mean the vessels disappear, small portions of some remain to give rise to adult structures



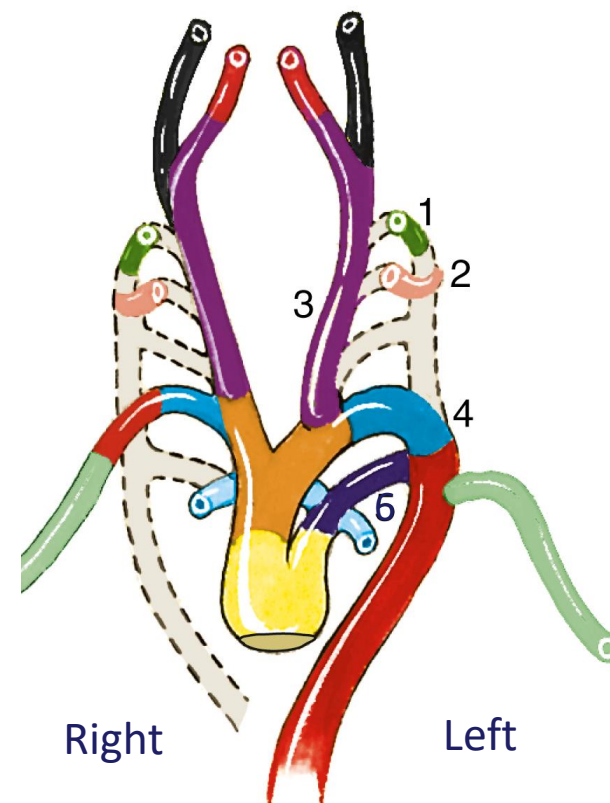
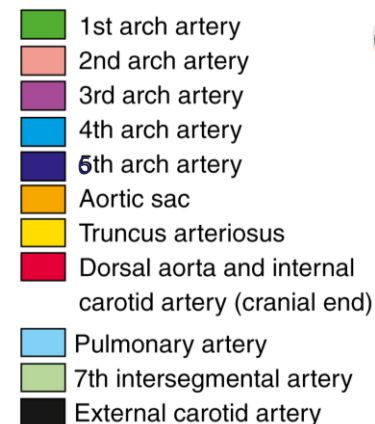
Week 4 & 5 - changes to dorsal aorta

- The portion of the dorsal aorta caudal to the 4th thoracic segment fuses together to form a **single median aorta**
- Day 35 - Dorsal aorta disappears between 3rd and 4th aortic arch arteries
 - Develops 3 sets of branches
 - Ventral (remnants of the vitelline arteries) to the gut and gut derivatives
 - Lateral to the retroperitoneal structures such as kidneys, adrenals and gonads
 - Dorsolateral intersegmental (intersegmental arteries) which penetrate the somites and give rise to vasculature of head, neck body wall, limbs and vertebral column
 - Become connected to the umbilical arteries

Modeling and remodeling of the aortic arch arteries and dorsal aortae occurs in weeks 5, 6 & 7

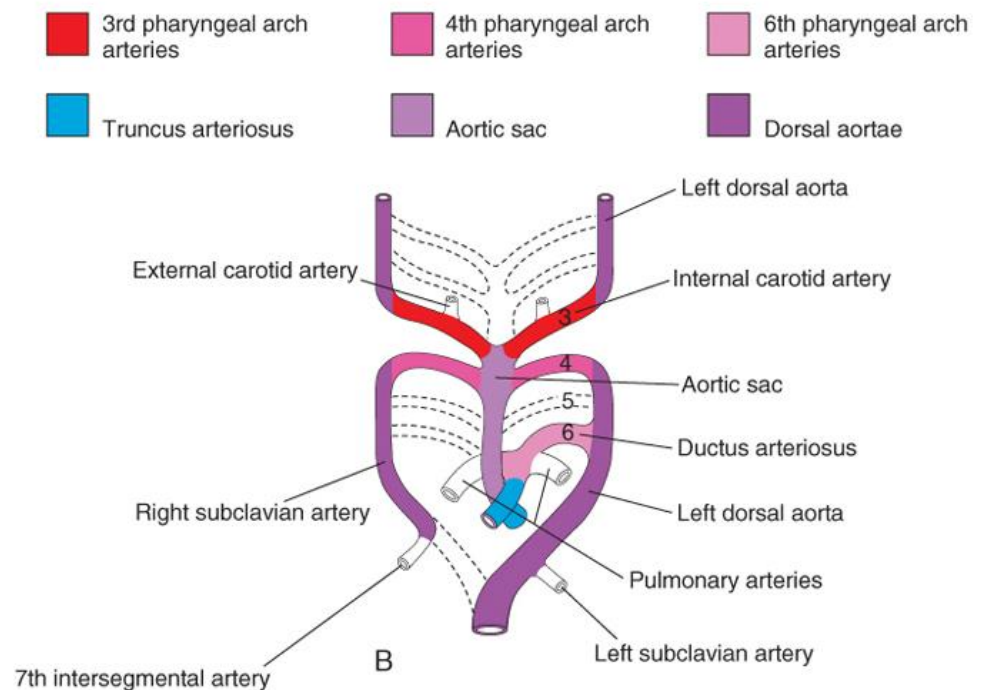
Week 7

- Right dorsal aorta loses connection with fused midline dorsal aorta and right 6th aortic arch artery
- 7th right intersegmental artery develops from the limb bud, joins remnant of dorsal aorta and 4th aortic arch artery = **right subclavian**
- Region of aortic sac connected to right 4th aortic arch artery is modified = **brachiocephalic artery**
- Left 4th aortic arch artery and a small portion of aortic sac = **aortic arch & proximal descending aorta**
- Left 7th segmental artery arises from left limb bud = **left subclavian**
- Left 6th aortic arch artery remains complete and becomes the **ductus arteriosus**



Adult derivatives of Aortic Arches

- **1st arch artery** largely disappears (~Day 27) (image)
 - Small part remains and forms the **maxillary artery**
- **2nd arch artery** later disappears (image)
 - Remaining parts are the **hyoid and stapedial arteries**
- **3rd arch**
 - **common carotid artery**
 - first part of the **internal carotid**



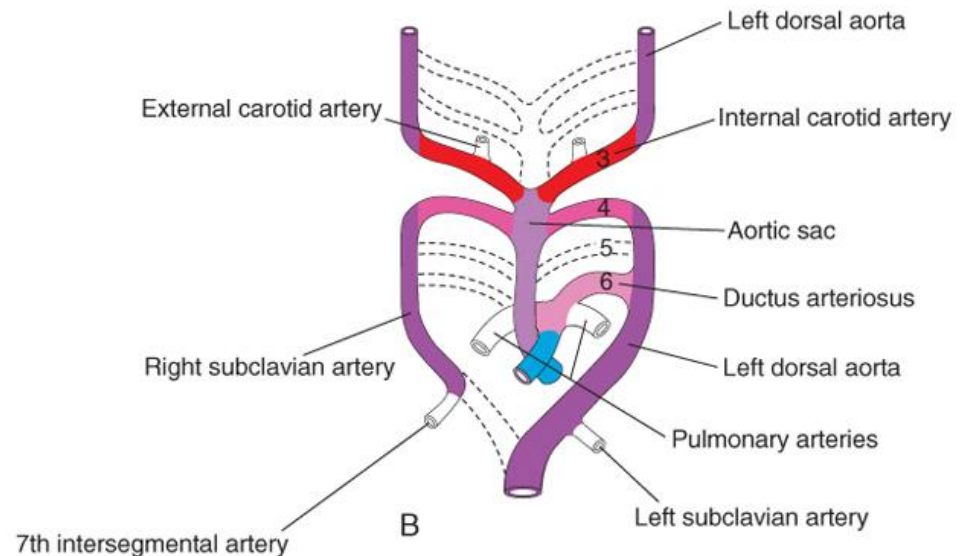
Adult derivatives of Aortic Arches

4th arch artery

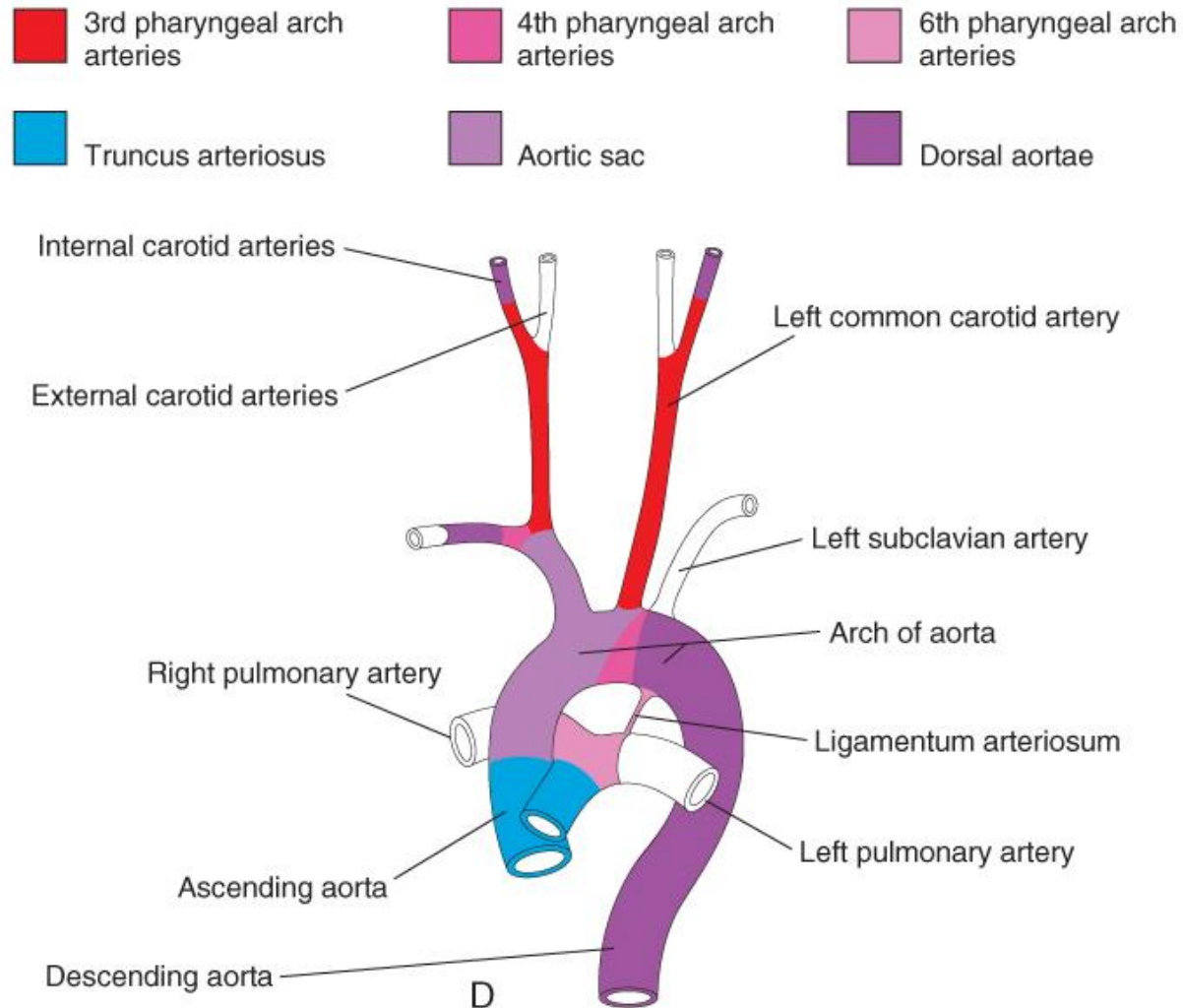
- on the left forms part of the **arch of the aorta**
- on the right forms the most proximal part of the **subclavian artery**

6th arch artery (pulmonary arch)

- On the right, proximal part becomes proximal segment of **right pulmonary artery**, the distal part degenerates
- On the left, proximal part becomes proximal segment of **left pulmonary artery**, distal part persists as the **ductus arteriosus**



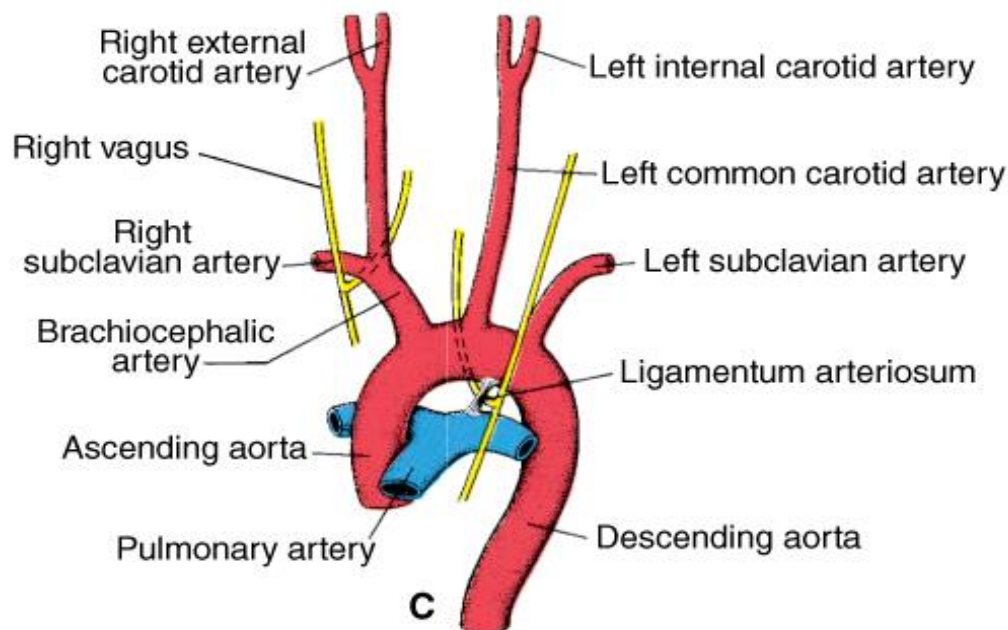
Summary



Sketch of the arterial vessels in a 6-month-old infant. The arteries that are not colored are not derived from the aortic arches.

Recurrent Laryngeal Nerves

- These nerves are branches of Vagus N. originally developed from the 6th pharyngeal arches
- As the heart descends, the nerves hook around the 6th aortic arches and ascend to the larynx
- This accounts for their recurrent course
 - On the right the 6th aortic arch artery disappears = hooks around the right subclavian artery
 - On the left the 6th aortic arch artery as the **ductus arteriosus (ligamentum arteriosum)** = hooks around the aortic arch





Aortic Arch Artery	Adult Derivatives
First arch artery	Regresses, Remnants form part of the maxillary arteries (supplies the ears, teeth, facial and eye muscles)
Second arch artery	Regresses, Remnants form the stapedial and hyoid arteries
Third arch artery	Proximal parts: common carotid Distal parts: join with the dorsal aortae to form the internal carotid arteries
Fourth arch artery	Left: part of the aortic arch Right: proximal part of the right subclavian
Fifth arch artery	Degenerates with no vascular derivatives or never forms
Sixth arch artery	Left: Proximal part: left pulmonary artery Distal part: ligamentum arteriosum Right: Proximal part: proximal part of the right pulmonary artery Distal part: degenerates

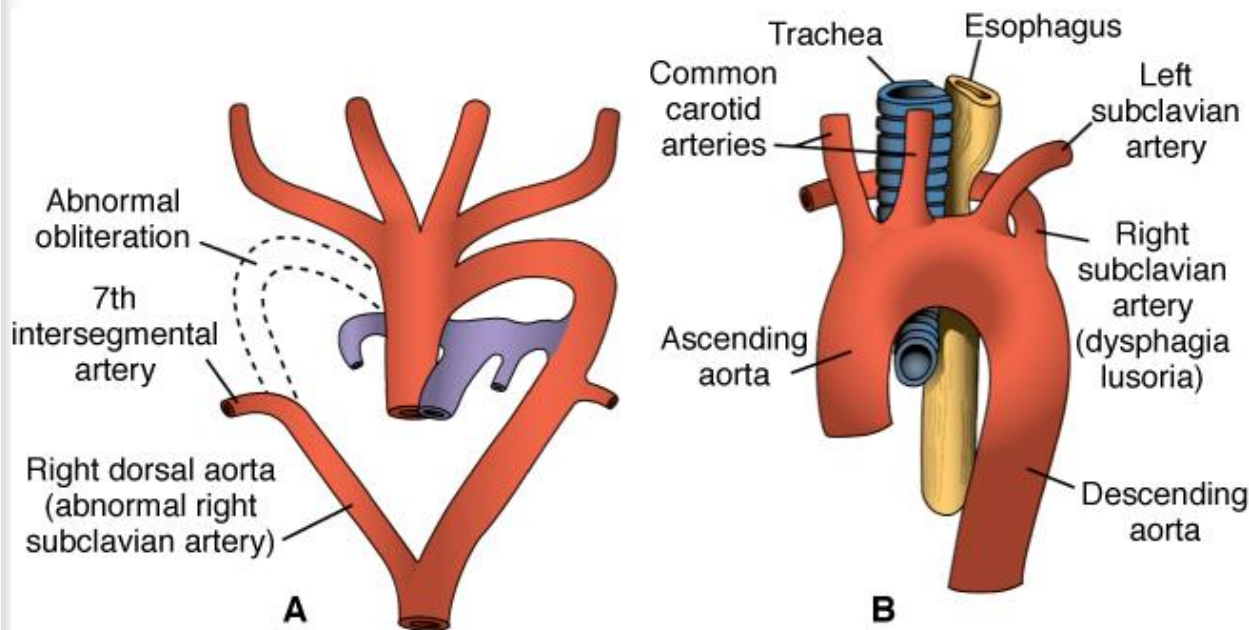
Embryological origin of the large arteries of the thorax

Aorta	Arch: Left 4 th arch artery, aortic sac and left dorsal aorta Ascending aorta: truncus arteriosus. Descending: Dorsal aorta
Carotid arteries	3 rd arch forms the common carotid and the proximal part of the internal carotid. The distal internal is from the dorsal aorta. External carotid is not an arch derivative, but the maxillary branch of it is.
Subclavian arteries	Right subclavian from the right 4 th pharyngeal arch artery, dorsal aorta, and the right 7 th cervical intersegmental artery Left subclavian from the Left 7 th cervical intersegmental artery and not an aortic arch derivative.
Proximal part of right and left main pulmonary arteries	Proximal part of both 6 th arch arteries.
Ductus arteriosus/ligamentum arteriosum	Distal part of Left 6 th arch artery.

Anomalous right subclavian artery

The anomalous artery crosses the midline behind the esophagus and can compress it, although this is not often tight enough to cause clinical symptoms

- Pathogenesis:
 - Obliteration of the right 4th aortic arch and the proximal part of right dorsal aorta
 - With persistence of distal part of right dorsal aorta



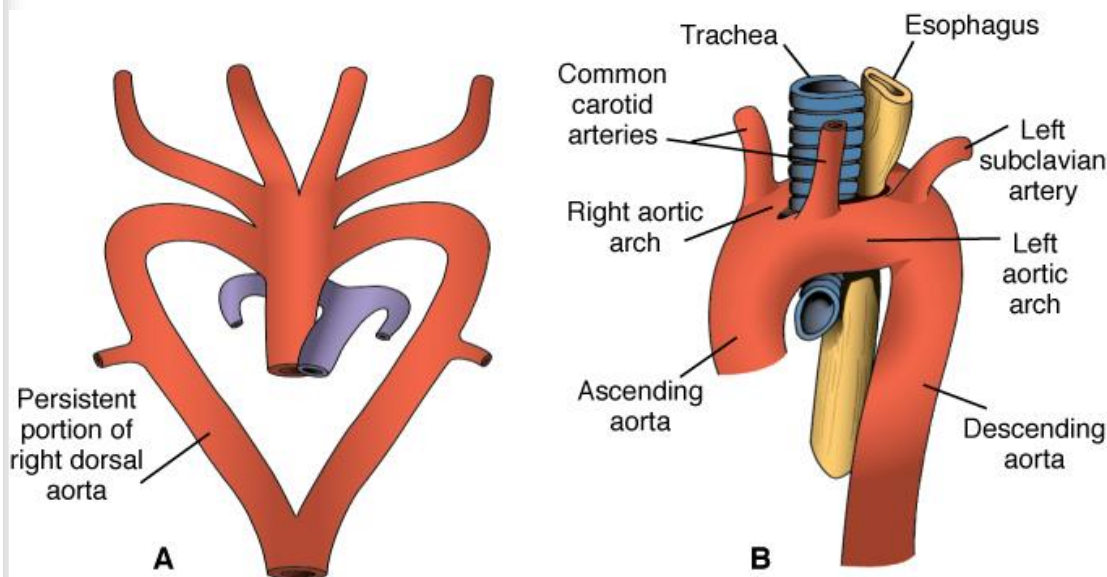
Normal development

Right 4th aortic arch artery - Forms proximal part of subclavian artery
Right dorsal aorta and right 7th intersegmental artery - Forms distal part of subclavian artery

Double aortic arch

- Pathogenesis:

- Persistence of right dorsal aorta between the origin of 7th intersegmental artery and its junction with left dorsal aorta (fails to disappear)**
 - A vascular ring surrounds the trachea and esophagus, compressing them and causing problems with breathing and swallowing

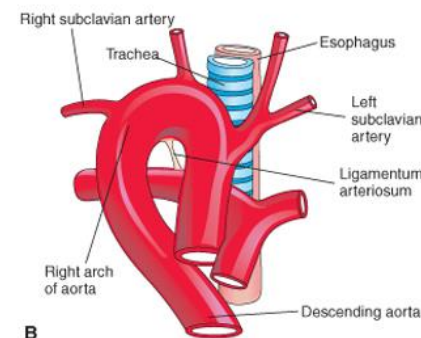
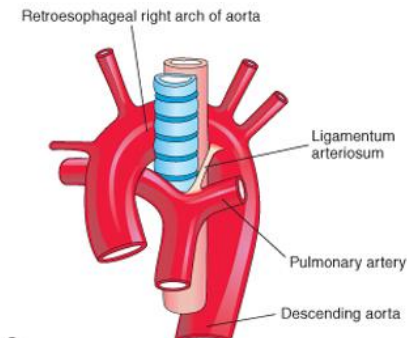


Normal development:
Proximal arch - Left 4th aortic arch and aortic sac
Distal arch- Left dorsal aorta

Right aortic arch

Two types

- With a retroesophageal component (Image C)
 - Original small arch of aorta involuted leaving right arch of aorta posterior to the esophagus
 - Ligamentum arteriosum attaches to the distal part of the arch and forms a ring which can compress trachea and esophagus
- Without a retroesophageal component (image B)
 - Ligamentum arteriosum passes from right pulmonary artery to right arch of aorta
- Pathogenesis:
 - **Persistence of the entire right dorsal aorta and abnormal involution of the distal part of the left dorsal aorta**
 - *The left subclavian is formed by the left 4th arch, left dorsal aorta and left intersegmental artery*



Coarctation of the Aorta (CoA)

- Congenital narrowing of a segment of aorta
- Male:Female (2:1)
- 70% associated with bicuspid aortic valve
- Present in 15-20 % of Turner syndrome patients
- Embryological theories:
 - abnormal involution of a small distal segment of the left dorsal aorta
 - muscle tissue of the ductus incorporates into the wall of aorta, which constricts when ductus closes at birth

